MIL-C-81582B
5 November 1975
SUPERSEDING
MIL-C-81582A(NAVY)
20 October 1969

MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, BAYONET COUPLING, ENVIRONMENT RESISTING, UMBILICAL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers environment-resisting, quick disconnect, EMI/RFI shielded and non-shielded umbilical, electric connectors and adapter assemblies with removable crimp or nonremovable solder-type contacts and accessories. Connectors are rated for operation from -55°C (-67°F) to 200°C (392°F). Adapter assemblies are rated for operation from -55°C (-67°F) to 125°C (257°F). The upper temperature is the maximum internal hot spot temperature resulting from any combination of electrical load and ambient temperature

1.2 Classification.

1.2.1 Classes. All classes shall be environment resistant and shall include the following features as specified in the applicable specification sheet

Class E - Grommet seal
Class F - Grommet seal and strain relief clamp
Class P - Potted seal
Class J - Insert seal with gland seal for jacketed cable

1.2.2 Contact designator. Contact style shall be designated by one of the following:

P - Pin contacts
S - Socket contacts

1 2.3 Military part number The military part number for qualified items, procured in accordance with this specification shall conform to the applicable specification sheet (see 3.1).

1.3 Accessories. This specification also covers the following accessories:

Adapters, Connector to Cable, Straight and 105°, open wiring and shielded cable.
Adapter, Connector to Connector, 90°
Connector, Dummy Stowage Receptacle
Caps, Protective

1 4 Insulation resistance Minimum insulation resistance versus hot-spot temperature is shown on figure 1.

1.5 Service life Service life is dependent upon hot spot temperature in accordance with figure 2

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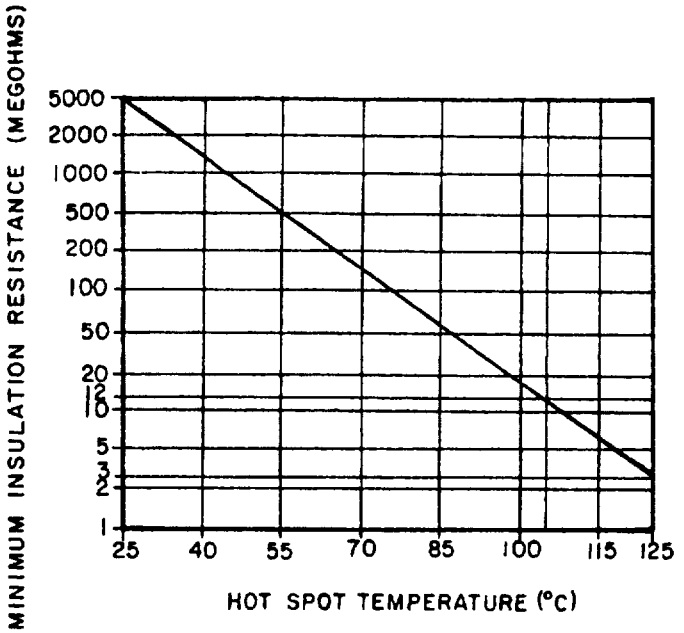


FIGURE 1A. Adapter assemblies.

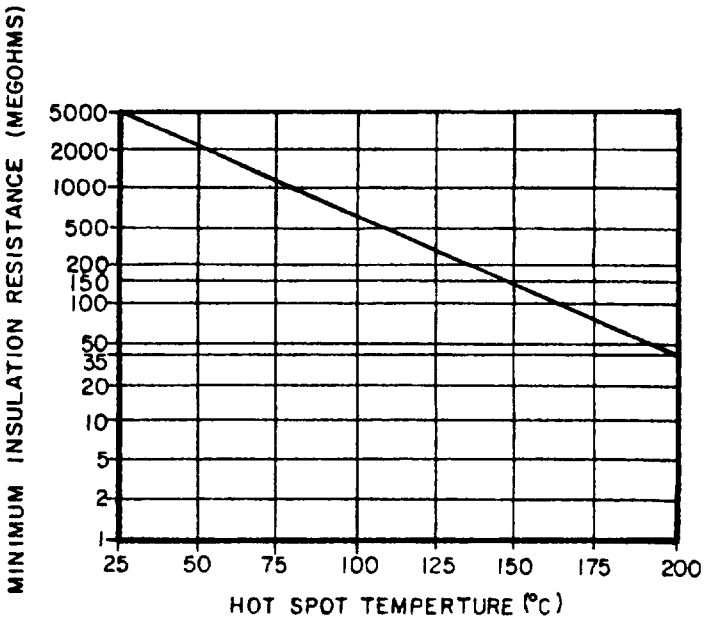


FIGURE 1B. Connectors.

FIGURE 1 Minimum insulation resistance vs hot spot temperature

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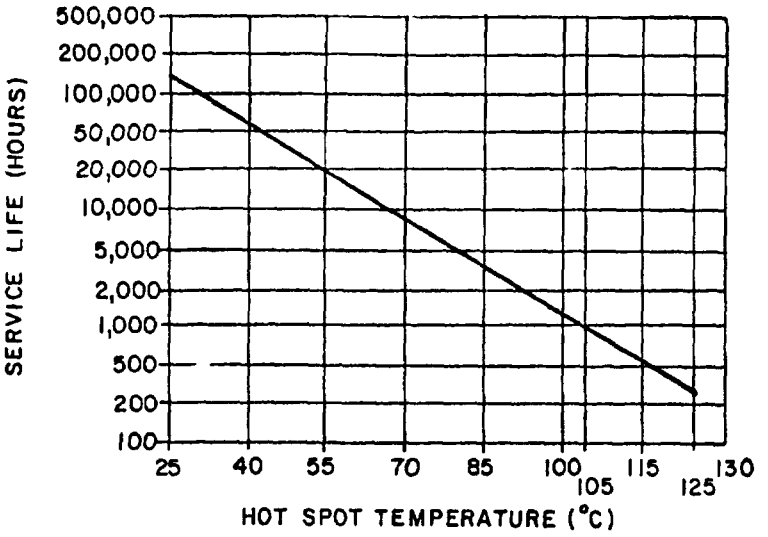


FIGURE 2A. Adapter assemblies.

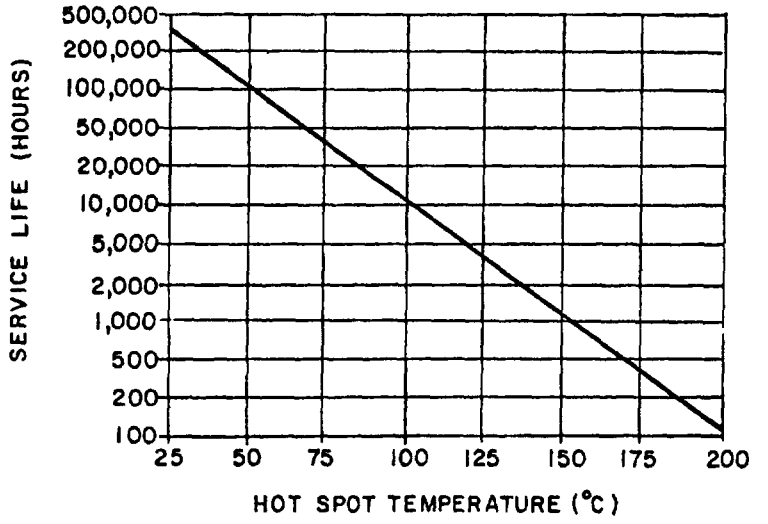


FIGURE 2B. Connectors

FIGURE 2 Service life versus hot spot temperature

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

L-P-395 - Plastic Molding (and Extrusion) (Nylon) Glass Fiber Reinforced
QQ-A-225 - Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes, Rolled, Drawn, or Cold Finished, General Specification for

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FEDERAL (Continued)

QQ-A-367 - Aluminum Alloy Forgings.
QQ-A-591 - Aluminum Alloy Die Castings.
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-764 - Steel Bar, Corrosion Resisting, Free Machining.

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-W-5086/1 - Wire, Electric, Hookup and Interconnecting, Polyvinyl Chloride Insulation, Nylon Jacket, Tin-Coated Copper Conductor, 600 Volt, 105°C
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-T-5624 - Turbine Fuel, Aviation Grades JP-4 and JP-5.
MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series, General Specification for.
MIL-S-8516 - Sealing Compound, Polysulfide Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-L-8937 - Lubricant, Solid Film, Heat Cured
MIL-T-10727 - Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-I-17214 - Indicator, Permeability. Low-Mu (Go-No-Go)
MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid
MIL-C-22520/1 - Crimping Tool, Contact, Electric, Hand for Contact Size 12 Through 20
MIL-C-22520/2 - Crimping Tool, Contact, Electric, Hand for Contact Size 20 Through 28.
MIL-W-22759/4 - Wire, Electric, Fluoropolymer-Insulated, TFE-Glass-FEP, Medium Weight, Silver-Coated Copper Conductor, 600 Volt
MIL-W-22759/5 - Wire, Electric, Fluoropolymer-Insulated, Abrasion Resistant, Extruded TFE, Silver-Coated Copper Conductor, 600 Volt
MIL-W-22759/7 - Wire, Electric, Fluoropolymer-Insulated, Abrasion Resistant, Extruded TFE, Medium Weight, Silver-Coated Copper Conductor, 600 Volt.
MIL-W-22759/9 - Wire, Electric, Fluoropolymer-Insulated, Extruded TFE, Silver-Coated Copper Conductor, 1,000 Volt
MIL-W-22759/11 - Wire, Electric, Fluoropolymer-Insulated, Extruded TFE, Silver-Coated Copper Conductor, 600 Volt
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
MIL-C-39029 - Contacts, Electric, General Specification for
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-C-45662 - Calibration System Requirements
MIL-C-55330 - Connectors, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods
MIL-STD-1285 - Marking of Electrical and Electronic Parts
MIL-STD-1344 - Test Methods for Electrical Connectors

(See Supplement 1 for list of associated Standards and Specification Sheets)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

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2 2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute Publications.
ANSIY14.5-1973 - Dimensioning and Tolerancing for Engineering Drawings.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

MIT Radiation Laboratory Series.
Volume 10, The Waveguide Handbook

(Application for copies should be addressed to the McGraw-Hill Book Co., Inc New York, New York.)

3. REQUIREMENTS

3 1 Specification sheets. All requirements herein apply to all the connectors and accessories covered by the applicable specification sheet. Detail requirements or exceptions applicable to the individual types shall be as specified herein or on the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3 2 Qualification. Connector assemblies and accessories furnished under this specification and the applicable specification sheet shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for the opening of bids (see 4 4 and 6.4).

3 2 1 Use of military identification. Military part identification shall not be applied to a product, except for qualification test samples (see 6 4), until notification has been received from the activity responsible for qualification that the product has been approved for listing on the Qualified Products List (QPL)

3 3 Materials Materials shall be as specified herein However, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification and the applicable specification sheet Acceptance or approval of any constituent material shall not be construed as a guaranty for acceptance of the finished product.

3.3.1 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metal in contact, which tends toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal spraying or metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. Dissimilar metals shall be as defined in 6.7 through 6.7.4 inclusive, and table XXI in hermetic seals, the 0.25 volt difference between the header material and the housing material is not applicable.

3.3.2 Nonmagnetic materials. The relative permeability of the connector assembly shall be less than 2.0, when measured with an indicator conforming to MIL-I-17214.

3.3.3 Contact materials. Contacts shall be made from suitably conductive materials protected from corrosion.

3.3.3.1 Contact plating. Contacts shall be gold plated to a thickness of 50 microinches (0.000050 inches) minimum in accordance with MIL-G-45204, type I, grade C, class I, over a nickel underplate in accordance with QQ-N-290, grade E. Accessory members (hood or spring, or both) shall be suitably protected from corrosion.

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3.3.4 Dielectric materials.

3.3.4.1 Insert and grommet. Insert and grommet materials shall be high grade dielectric having hardness, electrical, and mechanical characteristics suitable for the purpose intended.

3.3.4.1.1 Rigid dielectric material. Rigid dielectric materials shall be in accordance with L-P-395, MIL-M-14, or MIL-M-20693.

3.3.4.1.2 Resilient. Unless otherwise specified (see 3.1), the mating face of pin inserts shall be a resilient material within a Shore A Durometer range of 40 - 68

3.3.4.1.3 Insert faces. Insert faces shall be as specified in the applicable specification sheet (see 3.1).

3.3 5 Shells and metallic accessories material. Shells and metal accessories (see 1.3), shall be fabricated of high grade aluminum such as alloy 6061, temper T6 in accordance with QQ-A-367. Connectors shall have a grounding spring member as specified (see 3.1).

3 3 5 1 Component parts Other metal component parts may be fabricated from aluminum conforming to QQ-A-591 or QQ-A-225 or passivated stainless steel in accordance with QQ-S-764

3 3 5 2 Bayonet pins Bayonet pins shall be made of stainless steel

3 3 5 3 Finish Unless otherwise specified (see 3 1), aluminum parts shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3 A preliminary plating of another metal is permissible The resulting finish shall be olive drab (light to dark) and shall be electrically conductive.

3 3.5 4 Grounding spring member. The grounding spring shall be made of an electrically conductive metal suitable for the intended purpose.

3 4 Design and construction Connectors shall be of the design, construction, and physical dimensions specified (see 3.1) Connectors shall be so designed that neither the pins nor the sockets will be damaged during normal mating of counterpart connectors True position dimensions shown herein and on the specification sheets are in accordance with ANSIY14.5-1973.

3 4 1 Contact design. Contacts shall conform to MIL-C-39029, except as otherwise specified herein for solder contacts and thru-bulkhead contacts When specified in the contract or order, connectors may be supplied less crimp type contacts to permit procurement of these items in bulk.

3 4 1 1 Solder contacts Solder contacts shall be as specified (see 3 1), and shall be nonremovable from the insert Solder cups shall be such that during soldering no components will be damaged and no liquid solder shall escape

3 4 1 2 Crimp contacts. Crimp contacts shall be qualified as specified (see 3 1)

3 4.2 Insert design and construction Inserts shall be of voidless construction and shall be secured to prevent rotation within the shell The inserts shall be nonremovable from the shell and shall be installed in the position specified in the applicable specification sheet The insert shall be designed and constructed to eliminate all air paths between contacts by permanently attaching the dielectric members to each other The socket insert shall be designed to prevent a bent pin in a mating counterpart insert from penetrating the insert alongside the socket contact Inserts shall be so designed that MIL-S-8516 potting material will adhere to the insert without treatment by the user

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3.4.3 Shell design. Connector shells shall be seamless and shall retain their inserts in a positive manner

3.4.3.1 Screw threads. Screw threads shall conform to MIL-S-7742. Slight out-of-roundness beyond the specification tolerance is acceptable if threads can be checked without forcing the thread gages.

3 4 4 Coupling. Unless otherwise specified (see 3.1), the coupling type shall be a 3-pin bayonet design with a lanyard release mechanism Coupling shall be accomplished by clockwise rotation of the coupling ring; uncoupling shall be accomplished by the application of a force to the lanyard. The design shall also permit uncoupling by counterclockwise rotation of the coupling ring. The coupling rings shall be fluted or knurled to provide a gripping surface.

3.4.4.1 Engagement of connectors Counterpart connectors of any arrangement and accessories shall be capable of being fully engaged and disengaged without the use of tools. Engagement of connectors shall be defined as full insertion of pins into sockets and proper sealing of the mating insert faces Full engagement shall be indicated by an audible sound at the completion of the coupling cycle, and a positive detent shall be included in the coupling mechanism to lock connectors in the engaged position.

3 4 4 1.1 Bayonet pins Unless otherwise specified (see 3 1), bayonet pins on connector shells shall have surfaces whose color is yellow. The color shall be within the limits described in MIL-STD-104. The ends of the pins shall be visible through suitable holes in the coupling ring when the connectors are engaged and the coupling ring is in the locked position.

3.4 4 2 Shell polarization Polarization of connectors shall be accomplished by matched integral key(s) and keyway(s) of counterpart connectors The polarization of counterpart connectors shall take place before coupling rings are engaged, and before any pin contacts can touch the opposing insert face or sockets of the counterpart connector.

3 4 4 3 Engagement seal. Connectors shall contain sealing means so that engaged connectors comply with the requirements specified herein The design of the seal shall be such that in mated connectors all air paths between adjacent contacts and between contacts and shells are eliminated. There shall be interfacial mating of the engaged connector insert to provide dielectric under compression of 0.005 inch minimum

3.4 4 4 Lubrication. Bayonet coupling slots, including the inside diameter in the bayonet coupling slot area shall be coated with a lubricant conforming to MIL-L-8937, form B to aid in mating connectors. Features which are intended to provide potting compound anchorage shall be free of lubricant

3 4 4 5 Protective covers and stowage receptacles When mated to counterpart connectors, the protective covers and stowage receptacles shall maintain the connector free of moisture, prevent air leakage, and comply with the applicable requirements of table XVII. Protective covers and stowage receptacles shall be provided with an EMI/RFI shielding feature where required (see 3.1)

3 4 4 6 Lanyard Mounting of the lanyard to the connector shall be such that rotation of the connector for engagement will not shorten the effective length of the lanyard

3 4 5 Wire sealing.

3 4 5 1 Classes E and F connectors Classes E and F connectors shall be provided with a wire sealing resilient grommet and gland nut capable of sealing the wire sizes as specified in tables IA and IB.

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3.4.5.2 Class P connectors. Class P connectors shall be provided with a potting form suitable to accept and bond to MIL-S-8516 potting material Inserts of class P connectors shall be so designed that potting material will adhere to the shell and insert without treatment by the user

3.4.5.3 Class J connectors. Class J connectors shall be provided with a resilient gland and gland nut capable of sealing on appropriate single-jacketed multiconductor cables.

TABLE IA. Adapter assemblies - Wire range accommodations.

Wire barrel size	Wire size	OD of finished wire (inch) 1/		Wire number 2/ (ref)
		Crimp contact connectors	Solder contact connectors	
20	24	0.046 (min.)	0.060 (min.)	MIL-W-22759/7-24
	22	0.047 (min)	0 060 (min.)	MIL-W-22759/11-22, /7-22
	20	0.085 (max)	0.085 (max.)	MIL-W-22759/4-20
16	18	0.066 (min)	0 066 (min.)	MIL-W-22759/11-18
	16	0.109 (max.)	0.109 (max.)	MIL-W-22759/4-16
12	14	0 097 (min.)	0.097 (min.)	MIL-W-22759/9-14
	12	0.142 (max.)	0.142 (max.)	MIL-W-5086/1-12

1/ For OD's smaller than that specified, see 6.1(h).
2/ Wire numbers are for reference only and are not intended to restrict the use of acceptable wire-types with these adapter assemblies

TABLE IB Connectors - Wire range accommodations.

Wire barrel size	Wire size	OD of finished wire (inch)	Wire number (ref) 1/
20	24	0.041 min	MIL-W-22759/11-24
	22	0.047 min	MIL-W-22759/11-22
	20	0 090 max	MIL-W-22759/5-20
16	18	0 068 min.	MIL-W-22759/9-18, /11-18
	16	0 130 max.	MIL-W-22759/5-16
12	14	0 107 min	
	12	0 170 max	MIL-W-22759/5-12

1/ Wire numbers are for reference only and are not intended to restrict the use of acceptable wire-types with these connectors

3.4.5.4 Grommet sealing plugs The grommets of classes E and F connectors shall be designed to accept sealing plugs as specified (see 3.1), in lieu of wire where unused contacts are employed When specified in the contract or order, connectors may be supplied less grommet sealing plugs to permit procurement of these items in bulk

3.4.5.6 Grounding spring member When specified (see 3.1), the grounding spring member shall have a minimum of six fingers per inch The spring fingers shall be shaped in a manner which will ensure proper engagement of the mating shell. The grounding spring shall be permanently fixed to the shell periphery and the attachment of the grounding spring to the shell shall not impede spring flexing. The spring fingers shall engage the mating shell at a minimum of 0.090 inch prior to the pin-socket engagement

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3 5 Interchangeability. All connectors and accessories having the same military part number shall be completely interchangeable with each other with respect to installation (physical) and performance (function) as specified herein. Solder and crimp contact connectors shall be intermateable (see 6.1(f)).

3.6 Performance. Connectors shall perform as follows when subjected to the conditions and tests specified.

3.6 1 Contact protection, connector mating. Connectors equipped with pin contacts shall not permit the edge of the shell of the counterpart mating connector to touch the pin contacts (see 4.6.2).

3.6.2 Resistance to test probe damage (size 12 and smaller socket contacts). Socket contacts shall withstand the bending moment and depth of test probe insertion without evidence of visible damage (see 4.6 3) Contacts shall meet the requirements of 3.6.4 after test.

3.6.3 Contact insertion and removal force (crimp removable contact connectors only). Contacts shall meet the insertion and removal forces specified in table II, when tested as specified (see 4.6.4).

TABLE II Contact insertion and removal forces.

Contact size	Insertion and removal force (pounds max)
20 - 20	20
16 - 16	20
12 - 12	30

3.6.4 Contact engaging and separating force. The contact engaging and separating force shall not exceed the applicable values specified in table III (see 4 6.5)

TABLE III Contact engaging and separating forces (ounces).

Mating end size	Minimum separation force (ounces)	Maximum average engagement force (ounces)	Maximum engagement force (ounces)
20	0.7	12	18
16	2	24	30
12	3	24	30

3 6.5 Mating and unmating forces. The coupling torque for mating and unmating of counterpart connectors shall meet the torque requirements of table IV. The connectors used in this test shall have the complete complement of contacts. The dummy stowage receptacles and protective covers, when mated with their applicable connector, shall also comply with the torque requirement of table IV (see 4.6.6).

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TABLE IV. Mating and unmating forces.

Shell size	Torque (pound-inches)	
	Maximum mating and unmating forces	Minimum unmating force
8	8	1
10	12	1
12	16	2
14	20	4
16	24	4
18	28	4
20	32	6
22	50	20
24	60	25

3.6.6 Contact retention. Contacts shall be retained in their inserts when subjected to the contact retention test in accordance with the axial loads of table V. The axial displacement of crimp contacts shall not exceed 0 012 inch while under load with the accessory tightened or 0.015 inch while under load with the accessory removed. The axial displacement of solder contacts shall not exceed 0.012 inch after the load has been removed (see 4.6.7)

TABLE V. Contact retention with axial loads.

Contact size	Axial load (pounds-minimum)	
	Removable contacts	Nonremovable contacts
20 - 20	15	15
16 - 16	25	25
12 - 12	25	25

3.6.7 Thermal shock. There shall be no evidence of damage detrimental to the operation of the connector or assembly after being subjected to the applicable temperature extremes (see 3.1) of table VI in accordance with the thermal shock test (see 4.6.8).

TABLE VI Temperature extremes.

Test part	Extremes	Degrees C	Degrees F
Adapter assemblies	Low	$-55\ ^{+0}_{-3}$	$-67\ ^{+0}_{-5\ 5}$
	High	$+125\ ^{+3}_{-0}$	$+257\ ^{+5.5}_{-0}$
Individual connectors	Low	$-55\ ^{+0}_{-3}$	$-67\ ^{+0}_{-5\ 5}$
	High	$+200\ ^{+5}_{-0}$	$+392\ ^{+9}_{-0}$

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3 6.8 Dielectric withstanding voltage. Connectors shall show no evidence of breakdown or flashover when subjected to the test voltages and altitudes of table VII (see 4 6 9).

TABLE VII. Test voltage (ac, rms). 1/

Condition	Service rating I		Service rating II	
	Sea level	70,000 ft.	Sea level	70,000 ft.
Mated	1,500	1,000 2/ 350 3/	2,300	1,200 2/ 500 3/
Unmated	1,500	275	2,300	500

1/ These are not working voltages.
2/ Integral grommet or potted only.
3/ Removable grommet or no grommet.

3.6.9 EMI/RFI shielding effectiveness.

3 6.9.1 Shielding effectiveness. When specified (see 3.1), the shells shall exhibit a leakage attenuation equal to or greater than that specified in table VIII (see 4 6 10.1).

TABLE VIII EMI/RFI shielding effectiveness.

Frequency (MHz)	Leakage attenuation (dB)	Frequency (MHz)	Leakage attenuation (dB)
100	65	400	55
150	60	600	50
200	60	800	45
300	55	1,000	45

3 6 9 2 Shell to shell continuity. The overall DC resistance shall not exceed 0 005 ohms (see 4.6 10 2)

3.6.10 Fluid immersion Unmated connectors shall mate within the torque specified in table IV, after being subjected to fluid immersion (see 4.6.11).

3 6 11 Durability Counterpart connectors shall show no mechanical or electrical defects detrimental to the operation of the connector after 500 cycles of coupling and uncoupling (see 4 6.12).

3 6.12 Vibration Mated connectors shall not be damaged and there shall be no loosening of parts due to vibration Counterpart connectors shall be retained in full engagement and there shall be no interruption of electrical continuity longer than 1 microsecond (see 4 6.13).

3.6.13 Shock. Mated connectors shall not be damaged and there shall be no loosening of parts, nor shall there be an interruption of electrical continuity longer than 10 microseconds during the exposure to mechanical shock for adapter assemblies or 1 microsecond for individual connectors.

3 6 14 Acceleration Mated connectors shall not be damaged and there shall be no loosening of parts due to acceleration Counterpart connectors shall be retained in full engagement and there shall be no interruption of electrical continuity longer than 1 microsecond when tested in accordance with 4.6 15.

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3.6.15 Acoustic noise When specified (3 1), mated connectors shall not be damaged and there shall be no loosening of parts when exposed to acoustic noise. Counterpart connectors shall be retained in full engagement and there shall be no interruption of electrical continuity longer than 1 microsecond during exposure to acoustic noise (see 4.6 16).

3.6 16 Moisture resistance. Mated connectors, subjected to the moisture resistance test shall maintain an insulation resistance of 100 megohms or greater at 25°C for adapter assemblies and 1,000 megohms or greater at 25°C for individual connectors (see 4.6.17).

3 6.17 Insulation resistance. Insulation resistance shall be measured at ambient and elevated temperatures.

3.6.17.1 Insulation resistance at ambient temperature The insulation resistance at ambient temperature (25°C or 77°F) shall be greater than 5,000 megohms (see 4.6 18.1)

3.6.17.2 Insulation resistance at elevated temperature Table IX shows the minimum insulation resistance after exposure to the required conditions (see 4 6.18 2)

TABLE IX. Temperature.

Test part	Ambient temperature	Qualification inspection	Group B inspection	Insulation resistance megohms (minimum)
Adapter assemblies	105°C 125°C	1,000 hrs 250 hrs	720 hrs 250 hrs	12 3
Individual connectors	150°C 200°C	1,000 hrs 100 hrs	720 hrs 100 hrs	150 35

3 6.18 Salt spray (corrosion) Unmated connectors, stowage receptacles, protective covers, and accessories shall show no exposure of basis metal due to corrosion which will affect performance (see 4 6 19)

3 6.19 External bending moment. Connectors, stressed using the applicable bending moment shown in table X shall exhibit no evidence of damage as revealed by inspection with 3X magnification There shall be no interruption of electrical continuity (see 4 6 20)

TABLE X External bending moment

Shell size	Bending moment (inch-pounds)
10	200
12	300
14	350
16	419
18	433
20	437
22	440
24	450

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3.6.20 Temperature life. The connectors shall perform satisfactorily after exposure to the temperature life test and shall pass the succeeding tests in the qualification test sequence (see 4 6 21).

3 6 21 Ozone exposure The connectors shall show no evidence of cracking of materials or other damage due to ozone exposure that will adversely affect subsequent performance in the qualification test sequence (see 4 6.22)

3.6.22 Magnetic permeability. The relative magnetic permeability of connectors and accessories shall be less than 2 (see 4.6.23).

3 6.23 Insert retention. The insert shall retain its normal position in the connector shell for at least 5 seconds at maximum specified pressure (see 4.6.24)

3.6.24 Air leakage.

3 6 24.1 Connectors, nonremovable contacts and accessories The air leakage rate shall not be greater than 1 cubic inch per hour (4.55×10^{-3} cm^3/sec) (see 4.6.25.1)

3.6.24.2 Stowage receptacles and protective covers. The air leakage rate shall not be greater than 1 cubic inch per hour (4.55×10^{-3} cm^3/sec.) (see 4.6.25.3).

3.6.25 Altitude immersion connectors only). Unless otherwise specified (see 3.1), individual connectors shall have an insulation resistance of at least 5,000 megohms after being subjected to altitude immersion, and shall maintain a dielectric withstanding voltage at sea level as specified in table VII (see 4.6.26).

3.6 26 Contact resistance Contacts in the mated condition shall meet the contact resistance requirements of table XI in addition to the requirements of MIL-C-39029 (see 4.6.27)

TABLE XI. Contact resistance limits.

Contact size	Wire size	Test current (amperes)	Maximum voltage drop (millivolts)	
			Initial	After corrosion
20-20	20	7.5	55	65
	22	5 0	45	55
	24	3.0	45	55
16-16	16	13 0	50	60
	18	10.0	45	55
	20	7 5	45	55
12-12	12	23.0	50	60
	14	17 0	45	55

3.6.27 Cover chains. Protective covers with sash chains or wire rope as specified (see 3 1) shall withstand a 25-pound tensile test without damage (see 4.6.28)

3 6 28 Lanyard retention The lanyard, including all hardware and joints shall withstand an axial tensile force of 150-pounds minimum (see 4 6 29).

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3 6 29 Lanyard release force. The lanyard release force shall be less than 50 pounds at ambient temperature and less than 300 pounds at -55°C (see 4 6 30)

3 6 30 Coupling ring retention The connector shall withstand an axial separation force between coupling mechanism and plug shell when applied in both directions There shall be no loosening of component parts when subjected to the minimum force specified in table XII (see 4.6 31).

TABLE XII. Coupling ring retention forces (min)

Shell size	Force lbs.	Shell size	Force lbs.
8	45	18	75
10	45	20	95
12	45	22	110
14	45	24	130
16	60		

3 6 31 Low temperature high humidity. Mated wired connectors shall show no signs of deterioration or breakage and shall pass the lanyard release tests at -55°C (see 4 6 32).

3.6 32 Systems compatability Certification of compliance of systems compatability shall be furnished to the qualifying activity by the connector or accessory manufacturer However, certification to the requirements indicated herein does not relieve the manufacturer of responsibility for qualification testing of the product (see 4 4) Verification of systems compatability shall be conducted by the qualifying activity (see 4.6 33)

3.6.33 Stores release. Connectors and connector assemblies furnished under this specification and the applicable specification sheet shall be products which have been successfully installed and operated in the system for which the products are intended (see 4 6 33.1).

3 7 Identification of product Each connector and accessory shall be legibly and permanently marked on the shell or coupling ring in accordance with MIL-STD-130 and MIL-STD-1285 The part number shall be as shown (see 3 1).

3 7.1 Insert marking Inserts shall be marked as shown on the applicable specification sheet. Raised or depressed characters shall not be used on mating faces

3 7.1 1 Contact designations. Contact locations shall be designated by identifiable characters of contrasting color on the front and rear faces of the insert or insert assembly Positioning and arrangement of the characters shall be such that the appropriate contact cavity be readily identifiable On the rear face of solder contact inserts, those individual contact designations may be omitted where space limitations render identifiability marginal. The characters on any face of the connectors shall remain identifiable after completion of the tests specified in tables XIII through XVII

3.7.2 Grommet marking. Wire openings on the rear face of grommets shall be marked with legible characters of contrasting colors corresponding to the insert contact designators On grommets of solder contact connectors it is permissible to identify, where space limitations dictate, only those wire openings which are located on the vertical centerline

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3.8 Workmanship. Connectors and accessories shall meet all design dimensions and intermateability requirements of this specification. Loose contacts, poor molding fabrication, loose materials, defective bonding, damaged or improperly assembled contacts, peeling, or chipping or plating or finish, galling of mating parts, nicks and burrs of metal parts and post molding warpage will be considered adequate basis for rejection of items of quality inferior for the purpose intended

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. With the exception of the stores release requirements, the supplier is responsible for the performance of all inspection requirements as specified herein unless otherwise specified in the contract or purchase order. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 1 1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspection. The inspection of connectors shall be classified as follows

(a) Qualification inspection (see 4.4)
(b) Quality conformance inspection (see 4 5).
    (1) Inspection of product for delivery (see 4 5 1)
    (2) Inspection of preparation for delivery (see 4.5.2).

4 3 Inspection conditions and preparation of sample.

4 3.1 Inspection conditions Unless otherwise specified, all inspection shall be made in accordance with the general requirements of MIL-STD-1344

4 3 2 Preparation of samples Connectors (except for the systems compatability test), shall be wired with approximately 3 feet of wire, as applicable from table I A MIL-C-22520/1 or MIL-C-22520/2 crimping tool, as applicable, shall be used to crimp removable contacts to the wires. Soldering shall be in accordance with requirement 5 of MIL-STD-454 for nonremovable contacts.

4.4 Qualification inspection Qualification inspection shall consist of the examinations and test performed in the sequence in either table XIII or table XIV, and table XVII on the qualification test samples specified in 4 4 1 After receipt of the letter of authorization from the activity responsible for qualification (see 6.4), the applicant shall submit three copies of his test reports (certified by the Government inspector indicating the extent to which the tests were witnessed), together with the systems compatability certification and the required samples specified in 4.4 1 to the activity responsible for qualification.

4 4 1 Samples Test samples shall be produced with materials, equipment, and procedures normally used in production

4 4.1.1 Connector assemblies (classes E, F, and P) Five complete connector assemblies, class E wall-mount receptacles and class E or F straight plugs shall be provided. These assemblies shall have the most compact insert arrangements in each

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voltage and temperature range and in each shell size for which qualification is desired. Separate samples are required for crimp-contact connectors and solder-contact connectors Crimp contacts shall not be installed in connectors submitted for qualification testing. The wired samples of class P connectors shall be potted with a material suitable for the temperature conditions of 4.6.21.

(a) The connectors for test samples 1, 3, and 5 shall be assembled with wire approaching the applicable minimum diameter shown in table I.
(b) Connectors for test samples 2 and 4 shall be assembled with wire approaching the applicable maximum diameter shown in table I
(c) Qualification of these samples shall admit qualification of other types and the balance of insert patterns in classes E, F, and P by similarity If qualification of classes E and F is not sought, samples of class P shall be substituted for class E or F in the foregoing, except that nominal gage wire may be used. However, qualification of class P samples alone shall not admit qualification of classes E and F connectors.

4.4 1.2 Connector assemblies (class J). Two mated pairs of class J connector assemblies, in each shell size, shall be supplied.

(a) The connectors need not be wired but shall be assembled using a solid polychloroprene cylinder of suitable length and OD in accordance with table XVIII. The Shore A durometer of the test cylinder shall be from 75 to 85
(b) Qualification of these samples when subjected to the tests of Table XVII will admit qualification of all class J assemblies if classes E and F are being qualified at the same time or have previously been qualified to this specification. If not, class J assemblies shall be subjected to all of the tests of tables XIII and XIV.

4.4 1 3 Connector assembly for systems compatability. One complete connector assembly, in accordance with the applicable specification sheet, shall be provided for the systems compatability inspection

4.4 1 4 Shells for EMI/RFI testing. Two pairs of mating shells shall be selected for qualification and shall be tested without inserts or contacts installed These pairs of mating shells shall be submitted to the applicable qualification sequence specified in table XIII.

4.4.1 5 Contact retention feature samples A sufficient number of additional connectors shall be supplied to provide a minimum of 82 contact cavities of the size of contacts for which qualification of the crimp contact retention feature is desired

4 4 1 6 Protective covers Two protective covers of each shell size with mating classes E or F connectors, shall be subjected to the tests of table XVII for group B

4 4.1 7 Stowage receptacles. Two stowage receptacles of each size, with mating class E or F connectors, shall be subjected to the tests of table XVII

4 4 1 8 Adapters (cable to connector and connector to connector). Four adapters of each type shall be selected for qualification They shall be tested with a qualified connector or an accessory device having identical mating features Two of each type adapter selected shall be submitted to the applicable qualification inspection specified in table XVII The remaining untested samples shall be forwarded to the qualifying activity with the qualification test reports

4.4 2 Test routine. Sample units shall be subjected to the applicable inspection specified in tables XIII and XIV in the order shown Connector accessories shall be subjected to the inspection test sequence of table XVII in the order shown

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TABLE XIII. Qualification inspection for removable contact type connector.

Title	Requirement paragraph	Test paragraph	Test sample number						Mating shell
			1	2	3	4	5	6	1 & 2
xamination of product	3.1, 3 3 thru 3.5, 3.7, 3.8	4.6 1	X	X	X	X	X	X	X
MI/RFI shielding effectiveness	3.6.9.1	4 6.10.1							X
ontact protection, connector mating	3.6.1	4.6.2	X		X				
ontact resistance, initial	3 6 26	4.6.27	X	X					
robe damage	3.6.2	4.6 3	X						
ontact insertion and removable force	3.6.3	4 6 4	X		X				
oupling torque	3.6.5	4.6.6	X	X	X	X			
oupling ring retention	3.6.30	4.6 31	X						
ontact retention	3.6.6	4.6.7	X	X					
hermal shock	3.6 7	4.6 8	X	X					
ielectric withstanding voltage (mated at sea level)	3.6.8	4.6 9 1	X	X			X		
luid immersion (lubricating oil)	3.6.10	4 6 11		X					
luid immersion (hydraulic fluid)	3.6 10	4.6 11			X				
ibration	3 6.12	4.6 13	X						
hock	3.6.13	4 6.14	X						
cceleration	3 6.14	4.6.15			X				
coustic noise	3 6.15	4.6 16				X			
oisture resistance	3 6 16	4 6.17					X		
nsulation resistance (ambient temperature)	3.6 17.1	4.6.18.1	X	X	X	X	X		
alt spray (corrosion)	3 6 18	4.6.19					X		X
xternal bending moment	3 6 19	4.6 20					X		
emperature life	3 6.20	4 6.21	X						
nsulation resistance (elevated temperature)	3 6 17 2	4 6 18.2	X						
zone exposure (unmated)	3.6.21	4 6.22		X					
lagnetic permeability	3 6 22	4 6.23	X						
nsulation resistance (ambient temp)	3 6.17 1	4.6.18.1		X			X		
nsert retention	3 6 23	4.6.24	X	X	X	X			
ow temperature high humidity	3 6.31	4.6 32	X						
oupling torque	3 6 5	4 6.6	X	X	X	X			
oupling ring retention	3 6 30	4 6 31	X						
anyard release force	3.6.29	4 6 30			X				
ltitude immersion	3.6 25	4 6.26				X			
nsulation resistance (ambient temperature)	3.6 17.1	4 6.18 1				X			
ielectric withstanding voltage (mated at sea level)	3.6 8	4 6.9.1				X	X		
ontact retention	3 6 6	4.6 7	X	X	X	X			
ielectric withstanding voltage (mated at altitude)	3 6 8	4.6 9.2	X						
ielectric withstanding voltage (unmated at altitude)	3 6 8	4 6 9.2	X						
ontact insertion and removal force	3 6 3	4 6 4		X		X			
ontact resistance	3 6 26	4 6 27	X	X					
MI/RFI shielding effectiveness	3 6 9.1	4.6 10 1					X		X
urability	3.6.11	4.6.12				X			X
xamination of product	3 1, 3 3 thru 3 5, 3 7, 3 8	4.6 1	X	X	X	X	X		X
Systems compatability	3 6.32	4.6.33						X	
Stores release	3.6 32 1	4 6 33 1						X	

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TABLE XIV. Qualification inspection for nonremovable contact type connectors

Title	Requirement paragraph	Test paragraph	Test sample number							
			1	2	3	4	5	6	H-1	H-2
Examination of product	3 1, 3 3 thru 3 5, 3 7, 3.8	4 6 1	X	X	X	X	X	X	X	X
Contact protection, connector mating	3 6 1	4 6 2	X		X				X	X
Contact resistance, initial	3.6 26	4.6 27	X	X					X	
Contact engaging & separating force	3 6 4	4 6 5	X	X	X	X				
Probe damage	3 6 2	4 6 3	X		X	X				
Coupling torque	3 6 5	4 6 6	X		X	X				
Coupling ring retention	3 6.30	4 6 31	X							
Contact retention	3.6.6	4 6 7	X	X	X	X				
Thermal shock	3.6.7	4 6.8	X	X	X	X				X
Contact retention	3 6.6	4.6 7	X	X	X	X				
Dielectric withstanding voltage (mated at sea level)	3 6.8	4 6 9 1					X		X	X
Fluid immersion (lubricating oil)	3.6 10	4.6.11		X					X	
Fluid immersion (hydraulic fluid)	3.6 10	4.6.11			X				X	
Vibration	3.6.12	4 6.13	X						X	
Shock	3 6 13	4 6 14	X						X	
Acceleration	3.6 14	4 6 15			X				X	
Acoustic noise	3 6 15	4 6 16				X				X
Moisture resistance	3.6.16	4 6 17					X			X
Insulation resistance (ambient temperature)	3 6 17.1	4 6 18.1	X	X	X	X	X		X	X
Salt spray (corrosion)	3.6 18	4 6 19					X		X	
External bending moment	3 6 19	4 6 20					X			
Temperature life	3 6.20	4.6 21	X							X
Insulation resistance (elevated temperature)	3.6.17	4 6 18 2	X							X
Ozone exposure (unmated)	3 6 21	4 6 22		X						
Magnetic permeability	3 6 22	4 6 23	X							
Insulation resistance (ambient temperature)	3 6 17 1	4.6 18 1		X			X			
Insert retention	3.6 23	4.6.24	X	X	X	X				
Low temperature humidity	3 6 31	4 6.32	X							
Coupling torque	3.6 5	4.6.6	X	X	X	X			X	X
Lanyard release force	3.6 29	4.6 30			X					
Altitude immersion	3 6 25	4 6 26				X			X	
Insulation resistance (ambient temperature)	3 6.17 1	4 6 18 1				X			X	
Dielectric withstanding voltage (mated at sea level)	3 6 8	4 6 9 1				X	X			
Air leakage	3 6 24 1	4 6 25.1	X		X					
Contact retention	3 6.6	4 6 7	X	X	X	X				
Dielectric withstanding voltage (mated at altitude)	3.6.8	4 6 9 2	X							X
Dielectric withstanding voltage (unmated at altitude)	3.6.8	4 6.9 2	X							X
Contact resistance	3.6 26	4 6 27	X	X					X	
EMI/RFI shielding effectiveness	3.6 9	4 6 10	X			X				
Durability	3.6.11	4.6.12				X			X	
Examination of product	3 1, 3 3 thru 3 5, 3.7, 3 8	4 6 1	X	X	X	X	X		X	X
Systems compatability	3.6 32	4 6 33						X		
Stores release	3.6 32 1	4 6 33 1						X		

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TABLE XV Group B inspection for removable contact type connectors.

Title	Requirement paragraph	Test paragraph	Test sample number						Mating shell
			1	2	3	4	5	6	1 & 2
Examination of product	3 1, 3 3 thru 3.5, 3 7, 3 8	4 6.1	X	X	X	X	X	X	X
Contact insertion and removable force	3.6.3	4 6 4	X		X				
Coupling torque	3.6 5	4.6 6	X	X	X	X			
Coupling ring retention	3 6.30	4.6.31	X						
Contact retention	3.6 6	4 6.7	X	X					
Thermal shock	3.6.7	4.6.8	X	X					
Dielectric withstanding voltage (mated at sea level)	3.6.8	4.6.9.1	X	X			X		
Vibration (at -55°C)	3.6 12	4.6.13	X						
Shock	3 6.13	4.6 14	X						
Moisture resistance	3.6.16	4 6 17					X		
Insulation resistance (ambient temperature)	3.6.17 1	4.6.18 1	X	X	X	X	X		
Salt spray (corrosion)	3.6 18	4 6.19					X		X
External bending moment	3.6.19	4.6.20					X		
Temperature life	3.6.20	4.6 21	X						
Insulation resistance (elevated temperature)	3.6.17 2	4.6.18.2	X						
Insert retention	3.6.23	4.6.24	X	X	X	X			
Low temperature high humidity	3.6.31	4 6 32	X						
Coupling torque	3.6.5	4 6.6	X	X	X	X			
Coupling ring retention	3.6 30	4.6 31	X						
Lanyard release force	3.6.29	4.6.30			X				
Altitude immersion	3 6.25	4.6.26				X			
Insulation resistance (ambient temperature)	3 6.17.1	4.6.18.1				X			
Dielectric withstanding voltage (mated at sea level)	3.6.8	4 6.9.1				X	X		
Contact retention	3.6.6	4.6 7	X	X	X	X			
Dielectric withstanding voltage (mated at altitude)	3 6.8	4 6.9 2	X						
Dielectric withstanding voltage (unmated at altitude)	3 6.8	4 6 9 2	X						
Contact insertion and removal force	3.6 3	4 6 4		X		X			
Contact resistance	3 6.26	4 6.27	X	X					
Durability	3.6.11	4 6.12				X			X
Examination of product	3 1, 3 3 thru 3 5, 3 7, 3 8	4 6 1	X	X	X	X	X		X

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TABLE XVI. Group B inspection for nonremovable contact type connectors

Title	Requirement paragraph	Test paragraph	Test sample numbers							
			1	2	3	4	5	6	H-1	H-2
Examination of product - - - -	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X	X	X	X	X	X	X	X
Contact resistance, initial - -	3.6.26	4.6.27	X	X					X	
Contact engaging & separating force - - - - - - - - - - - -	3.6.4	4.6.5	X	X	X	X				
Probe damage - - - - - - - - -	3.6.2	4.6.3	X		X	X				
Coupling torque - - - - - - - -	3.6.5	4.6.6	X		X	X				
Coupling ring retention - - - -	3.6.30	4.6.31	X							
Contact retention - - - - - - -	3.6.6	4.6.7	X	X	X	X				
Thermal shock - - - - - - - - -	3.6.7	4.6.8	X	X	X	X				X
Contact retention - - - - - - -	3.6.6	4.6.7	X	X	X	X				
Dielectric withstanding voltage (mated at sea level)-	3.6.8	4.6.9.1					X		X	X
Vibration - - - - - - - - - - -	3.6.12	4.6.13	X						X	
Shock - - - - - - - - - - - - -	3.6.13	4.6.14	X						X	
Moisture resistance - - - - - -	3.6.16	4.6.17					X			X
Insulation resistance (ambient temperature) - - - - - - - -	3.6.17.1	4.6.18.1	X	X	X	X	X		X	X
Salt spray (corrosion)- - - - -	3.6.18	4.6.19					X		X	
External bending moment - - - -	3.6.19	4.6.20					X			
Temperature life - - - - - - -	3.6.20	4.6.21	X							X
Insulation resistance (elevated temperature) - - - - - - - -	3.6.17	4.6.18.2	X							X
Insert retention - - - - - - -	3.6.23	4.6.24	X	X	X	X				
Low temperature humidity - - -	3.6.31	4.6.32	X							
Coupling torque - - - - - - - -	3.6.5	4.6.6	X	X	X	X			X	X
Lanyard release force - - - - -	3.6.29	4.6.30			X					
Altitude immersion - - - - - -	3.6.25	4.6.26				X			X	
Insulation resistance (ambient temperature) - - - - - - - -	3.6.17	4.6.18.1				X			X	
Dielectric withstanding voltage (mated at sea level)- - - - -	3.6.8	4.6.9.1				X	X			
Air leakage - - - - - - - - - -	3.6.24.1	4.6.25.1	X		X					
Contact retention - - - - - - -	3.6.6	4.6.7	X	X	X	X				
Dielectric withstanding voltage (mated at altitude) - - - - -	3.6.8	4.6.9.2	X							X
Dielectric withstanding voltage (unmated at altitude) - - - -	3.6.8	4.6.9.2	X							X
Contact resistance - - - - - -	3.6.26	4.6.27	X	X					X	
Durability - - - - - - - - - -	3.6.11	4.6.12				X			X	
Examination of product - - - -	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X	X	X	X	X		X	X

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TABLE XVII. Qualification and group B inspection for connector accessories and crimp contact retention feature.

Title	Requirement paragraph	Test paragraph	Test sample 1	Test sample 2
Protective Covers, Stowage Receptacles and Cable Clamps				
Examination of product	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X X	X X
Magnetic permeability	3.6.22	4.6.23	X	X
Low temperature high humidity	3.6.31	4.6.32	X	X
Coupling torque	3.6.5	4.6.6	X	X
Vibration	3.6.12	4.6.13	X	X
Moisture resistance	3.6.16	4.6.17	X	X
Cover chains, tensile strength	3.6.27	4.6.28	X	X
Air leakage	3.6.24.1	4.6.25.1	X	X
EMI/RFI shielding effectiveness	3.6.9	4.6.10.2	X	X
Examination of product	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X	X
Crimp Contact Retention Feature				
Examination of product	3.1, 3.7, 3.8	4.6.1	82 Contact Cavities Minimum	
Contact insertion and removal force	3.6.3	4.6.4		
Contact retention	3.6.6	4.6.7.2		
Connector Assemblies, Class J				
Examination of product	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X X	X X
Magnetic permeability	3.6.22	4.6.23	X	X
External bending moment	3.6.19	4.6.20	X	X
Thermal shock	3.6.7	4.6.8	X	X
Vibration	3.6.12	4.6.13	X	X
Moisture resistance	3.6.16	4.6.17	X	X
Salt spray (corrosion)	3.6.18	4.6.19	X	X
Air leakage	3.6.24.1	4.6.25.1	X	X
EMI/RFI shielding effectiveness	3.6.9	4.6.10	X	X
Fluid immersion (lub. oil)	3.6.10	4.6.11	X	
Fluid immersion (hyd fluid)	3.6.10	4.6.11	X	
Examination of product	3.1, 3.3 thru 3.5, 3.7, 3.8	4.6.1	X	X

TABLE XVIII. Test cylinder OD sizes.

Shell size	Diameter ±0.016 inch
8	0.214
10	.250
12	.384
14	.462
16	596
18	646
20	.681
22	716
24	817

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4.4.3 Qualification rejection. There shall be no failures during any examination or tests of the connectors or accessories detrimental to the operation of the connector submitted for qualification tests After notification of any failure, detrimental to the operation of the connector, the activity responsible for qualification testing shall receive details of corrective action from the supplier before initiating any further tests deemed necessary to assure compliance with connector requirements

4.4 4 Retention of qualification. In order to retain qualification, a summary of group B test results shall be furnished at one or two year intervals (see 4.5.1 4 1), and shall cover the results of group B tests performed during that period The summary shall also include the number and type of any part failures and shall be forwarded via the Government Inspector to the activity responsible for qualification. If the summary of results indicates nonconformance with specification requirements, 4.5.1 4 2 (nonconformance) shall apply. Failure to furnish the summary shall result in loss of qualification for that product.

4.4.5 Assembly plants. Assembly plants must be listed on or approved for listing on the applicable qualified products list. The qualified connector supplier shall certify that the assembly plant is approved for the distribution of the supplier's parts The assembly plant shall use only piece parts supplied by the qualified connector supplier No testing other than visual examination is required of certified piece parts obtained from the qualified connector supplier, except when there is cause for rejection. All assemblies produced at the assembly plant shall be subjected to the quality assurance provisions specified herein. Quality control requirements, including Government inspection surveillance, shall be the same as required for the qualified connector supplier

4.5 Quality conformance inspection.

4 5 1 Inspection of product for delivery Inspection of product for delivery shall consist of groups A and B inspection.

4 5.1 1 Inspection lot. An inspection lot, as far as practicable, shall consist of all connectors covered by one specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4 5 1 2 Disposition of sample units Sample units which have been subjected to the group A inspection may be delivered on the contract or order Sample units which have been subjected to the group B inspection shall not be delivered on the contract or order

4.5 1 3 Group A inspection Group A inspection shall consist of the examinations and tests specified in table XIX and shall be made on the same set of sample units in the order shown. In process control of component parts, unrelated to lot sizes of finished connectors, may be used in lieu of examination of these components in the finished connectors to assure performance of these component parts

TABLE XIX. Group A inspection.

Examination or test	Connector type		Accessories	Requirement paragraph	Test par
	Removable	Non-removable			
Examination of product	X	X	X	3 1, 3 3 thru 3 5, 3 7, 3 8	4 6 1
Insulation resistance	X	X		3 6 17	4 6 18
Dielectric withstanding voltage (unmated)	X	X		3 6 8	4 6 9

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4 5 1 3.1 Sampling plan. Statistical sampling and inspection shall be in accord-ice with MIL-STD-105 for general inspection level II. The acceptable quality shall be .0.

4 5 1 3.2 Rejected lots If an inspection lot is rejected, the supplier shall ithdraw the lot, rework it to correct the defects, or screen out the defective units, s applicable, and reinspect. Such lots shall be separate from new lots, and shall be learly identified as reinspected lots. Rejected lots shall be reinspected using ightened inspection.

4 5.1 4 Group B inspection. Group B inspection shall consist of the applicable kaminations and tests specified in either table XV or table XVI, and table XVII. ipment shall not be held up pending results of this inspection

4.5.1 4.1 Sampling plan. Sample units of one size, insert configuration, and each lass representative of production at the time of selection shall be selected at 6-month ntervals. Upon passing this inspection, the supplier may select sample units every 2 months. If the second level of sampling is passed two successive times, the supplier ay select sample units every 24 months. In the event of a failure, sampling shall evert to the 6-month interval

4.5.1.4.2 Noncompliance. If a sample fails to pass group B inspection, the supplier hall take corrective action on the materials or processes, or both, as warranted, and n all units of product which can be corrected and which were manufactured under essen-ially the same conditions, with essentially the same materials, processes, etc., and hich are considered subject to the same failure. Acceptance of related lots shall be iscontinued until corrective action, acceptable to the Government, has been taken. fter the corrective action has been taken, group B inspection shall be repeated on new amples Group A inspection may be reinstituted, however, final acceptance shall be ithheld until the group B inspection has shown that the corrective action was success-ul In the event of failure after reinspection, information concerning the failure and orrective action taken shall be furnished the contracting officer and to the qualifying ctivity who will initiate action to remove the failing product from the qualified roducts list.

4 5 2 Inspection of preparation for delivery. Sample packages and packs and the nspection of the preservation-packaging, packing and marking for shipment and storage hall be in accordance with the requirements of MIL-C-55330

4 6 Methods of examination

4 6 1 Examination of product The connectors, accessories, and piece parts shall be xamined to insure conformance with this specification and the applicable specification heets In process control of component parts, unrelated to lot sizes of finished onnectors, may be utilized in lieu of examination of these components in the finished onnectors to assure conformance of these component parts

4 6.2 Contact protection, connector mating The front edge of the shell of the plug onnector shall be scooped as deep as possible within the front aperture made by the hell of the mating receptacle The front edges of both shells shall be in contact uring this test (see 3 6 1)

4 6 3 Resistance to test probe damage. One socket contact of each size shall be ested in accordance with method 2006 of MIL-STD-1344 The contact shall be installed in collet type holder or connector for this test Probe depths used shall be 1/2, 3/4 and ull depth (see 3 6.2)

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4.6.4 Contact insertion and removal force (removable crimp contacts only). Connectors shall be tested in accordance with method 2012 of MIL-STD-1344. Insertion and removal shall be done 9 times. Contacts shall not be wired.

4.6 5 Contact engaging and separating force. Socket contacts shall be tested in accordance with the contact engagement and separation test of MIL-C-39029. Socket contacts may be tested installed in the connectors (see 3.6.4).

4 6.6 Mating and unmating forces. Receptacles, plugs, dummy stowage receptacles, and protective covers shall be engaged with and disengaged from counterpart connectors in accordance with method 2013 of MIL-STD-1344. The torque shall be applied at a uniform rate of approximately one inch-pound per second (see 3.6.5).

4.6.7 Contact retention. Retention of contacts shall be tested in accordance with method 2007 of MIL-STD-1344 and table V One sample of each connector shall be tested. Measurement of displacement shall be done while maximum load is applied and shall meet the requirements of 3.6.6.

4 6.7.1 Crimp contacts. Contacts shall be tested as specified in 4.6.7. Rear accessory hardware shall be connected and tightened on connector, the axial load shall be applied in the outward direction from the mating end of contacts only (see 3.6.6).

4.6.7 2 Crimp contact retention feature. Connectors shall be tested as specified in 4.6.7.1, except that the rear accessory hardware shall be removed prior to the test and the loads shall be applied successively to the mating and wire barrel ends of the contacts. The contacts selected for this test shall be crimped to metal rods complying with figure 3. The rods shall be capable of being crimped to the contacts with the MIL-C-22520/1 or MIL-C-22520/2 tool and shall be capable of being used to apply the specified axial load, without excessive bending, to the rear of the contacts during this test. Contact cavities not filled with rod contact assemblies shall be filled with wired contacts (see 3.6.6).

4.6.7 3 Solder type. Contacts shall be tested as specified in 4.6 7 The load shall be applied only to the mating ends of contacts and the grommet retaining nut shall be loose Displacement shall be measured after the specified load is removed (see 3 6 6)

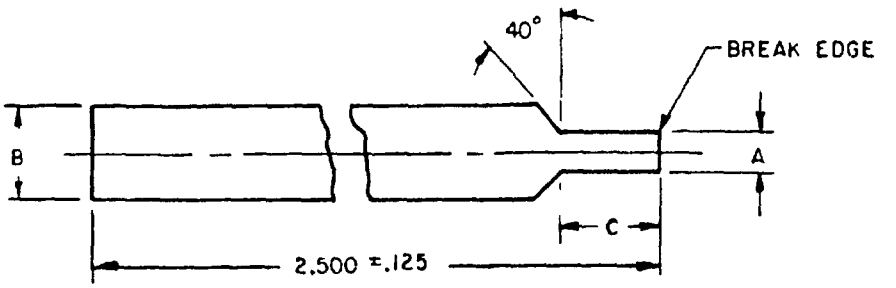
4 6.8 Thermal shock. Unmated connectors or unmated connectors with adapters shall be tested in accordance with Method 1003 of MIL-STD-1344, except that during the low temperature extreme of the fifth cycle, while at a minimum (-55°C) ambient, the connectors shall be mated and unmated 5 times At the completion of the last cycle, the connectors shall be returned to room temperature for further inspection Test condition B shall be used for connectors and test condition C shall be used for adapter assemblies (see 3.6.7).

4 6 9 Dielectric withstanding voltage (see 3.6.8)

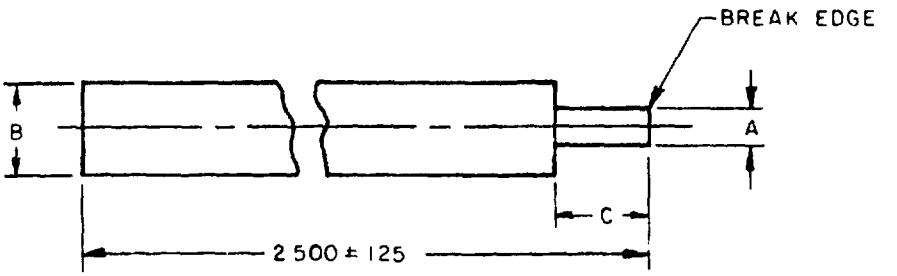
4.6.9 1 Dielectric withstanding voltage (sea level) Mated and unmated connectors shall be tested in accordance with Method 3001 of MIL-STD-1344. The applicable test voltages (see table VII) shall be applied between all adjacent contacts and between the shell and each contact closest to the shell If an insert possesses more than one service rating, similar connections shall be made for the different test voltages as necessary (see 3 6.8).

4 6 9.2 Dielectric withstanding voltage (altitude) The connectors shall be tested in accordance with Method 105, test condition C, MIL-STD-202 After 30 minutes at the simulated altitude, the connectors shall be tested as specified in 4 6 9 1 (see 3 6 8).

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ROD FOR SIZE 20 CONTACT



ROD FOR SIZE 16 AND 12 CONTACTS

CONTACT SIZE	A DIA	B DIA	C ± 010
20	047 045	080 075	187
16	065 063	100 .095	294
12	097 095	142 137	294

NOTE
All diameters to be concentric with each other within 004 TIR.

FIGURE 3 Rod for retention feature test

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4 6 10 EMI/RFI shielding effectiveness (see 3 6 9)

4 6 10 1 Shielding effectiveness. The electromagnetic susceptibility of mated pairs of shells shall be measured in a tri-axial radio frequency leakage tester as shown on figure 4 The RF leakage from the coaxial line through the shells of the connector pairs into outer coaxial geometry shall be measured at the frequency shown in table VIII (accuracy of frequency ±5 percent) The level of detected signal power indicated by a tuned radio frequency field intensity meter is a result of RF leakage from the mated pair of shells The test set-up and procedure shall be as shown on figure 5 (see 3.6.9.1)

4 6 10.2 Shell to shell continuity The dc resistance of the wired, mated, assembled connectors shall be measured between the points shown on figure 5. The applied potential shall be 1.5 volts dc maximum. A resistance shall be inserted in the circuit to limit the current to 0 100 ±0.010 amperes. Probes with spherical ends of 0.05 inch minimum shall be used to make the voltage measurements and shall be placed on the simulated panel and at the extreme edge of the plug as shown on figure 6 The probe shall not puncture or otherwise damage the connector (see 3 6 9 2)

4 6 11 Fluid immersion. Unmated connectors shall be immersed fully in the fluids specified below for the required periods. At least one connector shall be immersed in each fluid. After removal from the fluid, each connector shall remain for 1 hour in free air at room conditions Subsequent testing shall be performed on connectors mated with mating connectors that had been immersed simultaneously When required, other fluids may be specified (see 3.1 and 3 6 10)

(a) Hydraulic fluid conforming to MIL-H-5606 - 20 hours
(b) Lubricating oil conforming to MIL-L-23699 - 20 hours
(c) Turbine fuel (JP-5) conforming to MIL-T-5624-5 minutes at 120°C ±3°C

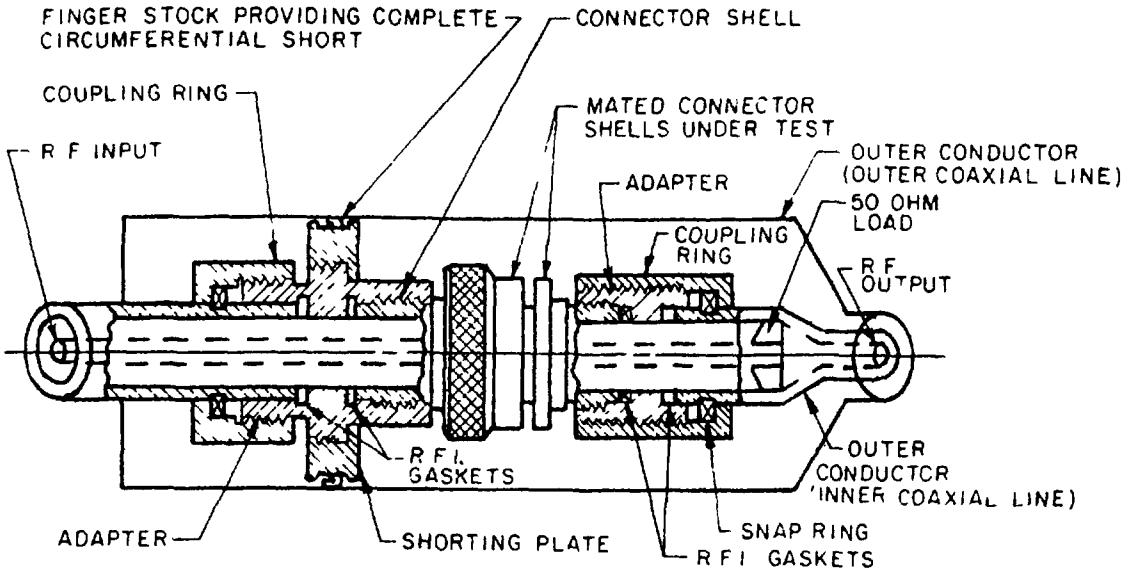
4.6 12 Durability. Counterpart connectors shall be mated and unmated at a maximum rate of 300 cycles per hour in a manner to simulate actual service in that plugs and receptacles shall be completely separated during each cycle (see 3 6 11)

4 6.12 1 Uncoupling (coupling ring) Connectors shall be mated and unmated 300 times with the coupling rings

4.6 12.2 Uncoupling (lanyard release) Connectors shall be mated and unmated 200 times using the lanyard release The connector shall be rigidly mounted as in service. The direction of force application shall be at an angle of 15° from the connector longitudinal axis for the first 50 cycles, along the longitudinal axis for the next 100 cycle cycles and 15° from the connector longitudinal axis on the opposite side for the last 50 cycles. The force shall be applied at maximum rate of 50 pounds per second

4 6 13 Vibration. Each receptacle shall be mounted on a suitable fixture, which in turn shall be attached to a vibration table. A suitable sensor shall monitor the vibration of the receptacle at a point on or near the receptacle. A counterpart plug shall be engaged with the receptacle and held by normal locking means without the use of safety wire. The wire bundles shall be clamped to nonvibrating points at least 8 inches from the rear of the connectors The wire bundle shall be defined as a complete compliment of size 20 wire, one braided cable shield, and an outer jacket of an acceptable insulation material to meet the temperature requirements. The average cable weight shall be 0 4 oz per inch minimum and shall have a minimum length of 24 inches The connector assembly shall be tested in accordance with method 214, test condition I, test condition letter A, of MIL-STD-202 All contacts shall be wired in series, and a current of 100 10 milliamperes (mA) shall flow through the series circuit during the test A suitable instrument shall monitor the current flow and indicate any discontinuity longer than 10 microseconds The assembly shall be vibrated 8 hours in the longitudinal direction and 8 hours in the perpendicular direction (See 3 6 12)

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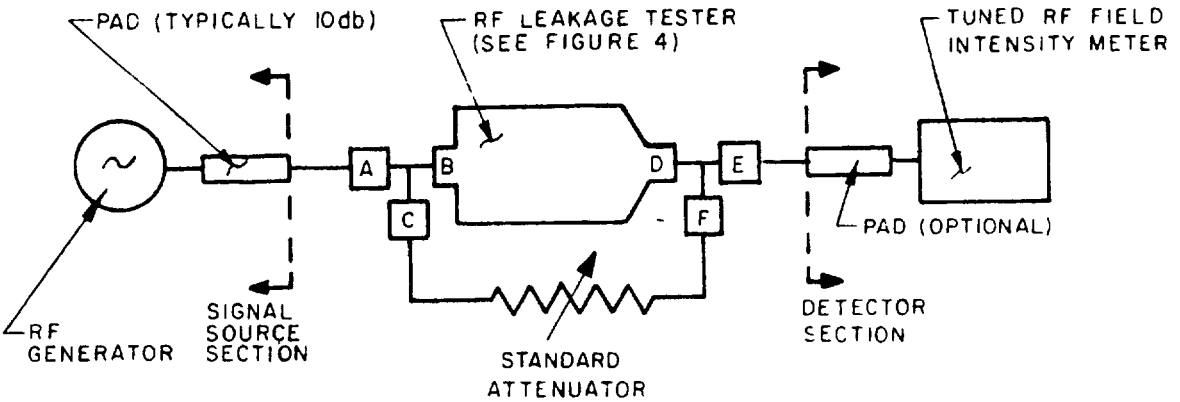
The mated pairs of connector shells shall be connected to the center coaxial line as shown and tested as shown in fig 5. Attachment of the connector shells to the outer conductor of the inner coaxial line shall be accomplished using R F gaskets to prevent leakage through the adapters

TESTER DESIGN REQUIREMENTS AND TERMINATIONS

a The characteristic impedances of the coaxial lines comprising the leakage tester shall be a nominal 50 ohms
b The VSWR (Zo = 50 ohms) looking into B shall be 1 5 or less when the mated pairs of connector shells are removed and replaced by a 50 ohm line section
c Existance of higher order modes, especially the transverse electric 1 1 (TE_{11}) mode will cause errors in required measurements The frequencies at which these modes exist must be avoided The frequencies may be calculated by referring to Section 2.4b of The Waveguide Handbook, Vol. 10 of the M.I.T. Radiation Laboratory Series, when the diameters of the outer coaxial section are known
d The input and output VSWR of the standard attenuator (see fig 6) should be less than 1 5 in the 20 to 100 dB attenuation range
e The output impedance of the signal source and the input impedance of the detector (see fig 5) shall be a nominal 50 ohms having a maximum allowable VSWR of 1 5
f Connectors A B C D E. and F (see fig 6) should be of a low leakage type which exhibit R F leakage attenuation of ~100 dB

FIGURE 4. Tri-axial RF leakage tester

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A standard attenuator capable of providing at least 100 dB (accuracy ± 3 dB) of attenuation in one dB steps shall be used to measure the RF leakage

1 Tune the detector to the generator signal frequency by connecting C to A and F to E, using approximately 85 dB attenuation in the standard attenuator
2 Disconnect C from A and F from E and couple A to B and D to E Adjust frequency slightly, if necessary, to maximize output indication on the tuned Radio Field Intensity Meter Record output level
3 Disconnect A from B and D from E Connect C to A and F to E within one minute, so as to prevent excessive drift of voltage in the signal source and of calibration in the Tuned Radio Frequency Field Intensity Meter Adjust the standard attenuator until the output level recorded in Step 2 is reached The value of the attenuation provided by the standard attenuator is a measure of the RF leakage attenuation of the mated pair of shells, measured in dB of power

FIGURE 5. Test set-up and procedure for shielding susceptibility test

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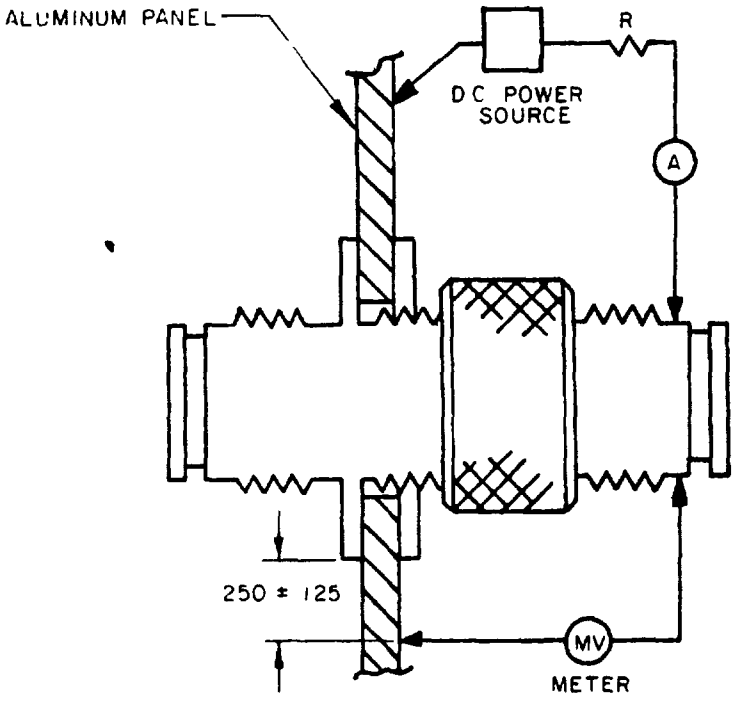


FIGURE 6. Shell to shell continuity test diagram.

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4.6.14 Shock Mated connector shall be tested in accordance with method 2004, test condition A, of MIL-STD-1344, except only one shock instead of three shall be applied in each direction (6 shocks). Receptacles shall be mounted similar to 4.6 13 1 Plugs shall be engaged with the receptacles and held by normal locking means only All contacts shall be wired in series and the wire bundles or cables shall be clamped to structures that move with the connectors. A minimum of 8 inches of wire or cable shall be unsupported behind the rear of each connector. A suitable instrument shall be employed to indicate any discontinuity or interruption of current flow (see 3 6.13)

4.6.15 Acceleration Connectors shall be tested in accordance with Method 2011, test condition A, of MIL-STD-1344. Contacts shall be wire to check for continuity during test (see 3.6.14)

4.6 16 Acoustic noise. The mated and wired connectors shall be mounted as in normal service to a rigid plane and tested in accordance with Method 515 of MIL-STD-810. The noise level and duration of exposure shall be category C for 30 minutes. At least 100 milliamperes shall flow through the series wired contacts during the test. A suitable instrument shall be employed to monitor the current flow and to indicate any discontinuity of contact or interruption of current flow (see 3.6.15).

4 6 17 Moisture resistance Moisture resistance test specimens shall be subjected to the high humidity (see 4 6.17.1), or extreme humidity range (see 4 6.17 2) moisture tests, as applicable. The connectors shall be wired and mated to the counterpart connectors. They shall be mounted horizontally with the wires descending into the backshell with no drip loops or splices within the chamber The wires shall leave the chamber through vaportight seals Connectors without rear seal grommets shall have their terminals suitably protected Prior to the beginning of the test and at the end of the test period and while at the high humidity, the insulation resistance between each contact and other contacts shall be determined as specified in 4.6 18 (see 3 6 16).

4 6.17 1 Moisture resistance at high humidity (crimp contact connectors) Mated crimp contact connectors shall be tested in accordance with Method 1002 1, Type II test of MIL-STD-1344 (except for vibration)

4.6.17 2 Moisture resistance, extreme humidity range (solder contact connectors). Mated solder contact connectors shall be subjected to the following test. The test chamber shall consist of a box approximately 12 inches wide by 24 inches long, capable of being sealed. A suitable open screen tray shall be provided to support the test specimens approximately 8 inches below the top of the box Provisions shall be made to bring out wires for measurement purposes through vaportight seals near the top of the box Suitable controls shall be provided that will cause the chamber air temperature to drop 5°C (9°F) once each hour for 20 days, from any temperature between 22° and 28°C (72° and 82°F), causing heavy condensation to form on the samples. The bottom of the test chamber shall be covered with approximately 1/4 inch of tap water to start the test The heat application to supply the temperature variation shall be radiation in nature and shall be applied to the underside of the test chamber

4 6 18 Insulation resistance (see 3 6 17)

4 6 18 1 Insulation resistance at ambient temperature Unmated connectors shall be tested in accordance with Method 3003 of MIL-STD-1344. The contacts selected shall be those having the closest spacing between measurement points (see 3 6 17 1)

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4 6.18.2 Insulation resistance at elevated temperature. The insulation resistance shall be measured in accordance with 4.6.18.1, except that the connectors shall have been exposed to an ambient temperature shown in table IX The measurements shall be made at the end of the temperature life (see 4.6.21), and while the connectors are at the elevated temperature

4 6.19 Salt spray (corrosion). Unmated connectors, shells, and accessories shall be subjected to a salt spray test in accordance with Method 1001, test condition B of MIL-STD-1344. The salt concentration shall be 5 percent. The specimens shall then be dried in a circulating air oven at a temperature of 38° ±3°C (100° ±5°F) for a period of not less than 12 hours, after which they shall be removed and inspected. Prior to all subsequent tests, corrosion tested connectors shall be engaged and disengaged for one cycle to remove free salt deposits (see 3.6.18).

4 6 20 External bending moment The receptacle connector shall be rigidly mounted as in normal service. Before mating the plug connector to the receptacle, the "J" adapter or test torque arm shall be attached as shown on figure 7. After mating the plug and receptacle connectors, the distance "L" from the point of load application "P", to the mounting panel shall be determined. The load to be applied at point "P" shall then be determined as the bending moment listed in table X divided by the lever arm "L". This load shall be applied at a rate of approximately 10 pounds per second until the required load is achieved. The load so applied shall be held for a period of 1 minute after which the load shall be released. Continuity of the contacts shall be monitored during the test. The connector shall be removed, unmated, and inspected to 3X power magnification for damage or breakage (see 3 6.19).

4 6.21 Temperature life. The connectors shall be tested in accordance with Method 1005 of MIL-STD-1344, except that the applicable ambient temperature and duration of test shall be as specified in table IX. Following these tests, and at the temperature specified, the insulation resistance shall be measured (see 1 4 and 3 6.20).

4 6 22 Ozone exposure. The unmated connectors shall be tested in accordance with test method 1007 of MIL-STD-1344. At the end of the test, the samples shall meet the specified requirements (see 3.6 21).

4 6 23 Magnetic permeability. Permeability shall be checked with an instrument conforming to MIL-I-17214 The connectors may be wired or unwired, as convenient, but shall not be carrying current (see 3.6 22)

4 6 24 Insert retention Inserts in wired connectors less removable grommets shall be tested in accordance with test Method 2010 of MIL-STD-1344. A force of 75 pounds per square inch shall be applied, and insert shall meet the requirements of 3.6.23

4 6.25 Air leakage

4.6 25 1 Connectors with solder contacts and environmental sealing adapters. Connectors with solder contacts and sealing adapters shall be mounted in a manner suitable for application of a 30 psi pressure differential across the connectors. The leakage rates shall be measured in both directions after 30 minutes of exposure to the low temperature extremes of table VI and while at the low temperature (see 3.6.24 1)

4 6 25 2 Stowage receptacles and protective covers Stowage receptacles and protective covers shall be mated to connectors having either contacts or inserts removed so that a pressure of 10 psi can be applied against the insides of the protective covers or stowage receptacles and tested in accordance with method 112, test condition C of MIL-STD-202 (see 3 6 24 2)

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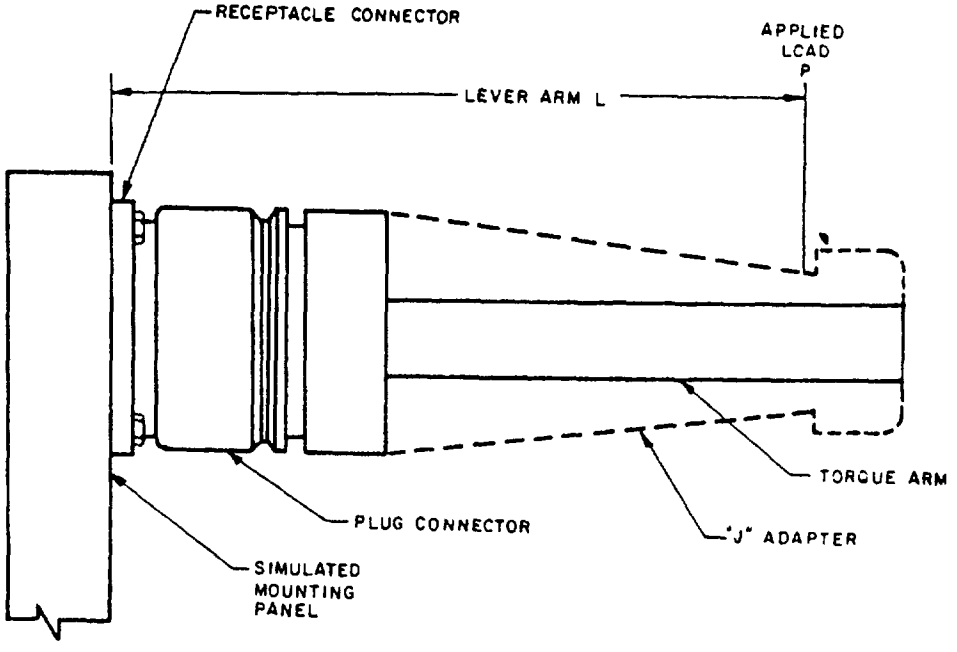


FIGURE 7. External bending moment test set-up.

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4.6 26 Altitude immersion (individual connectors only) Mated connectors shall be ested in accordance with test Method 1004 of MIL-STD-1344, except that water tempera- ure shall be approximately 30°C Wire ends, opposite the connector, shall be terminated ithin the altitude chamber, shall be exposed to the chamber atmosphere but not submerged, nd shall not be sealed. Insulation resistance requirements shall be as specified in .6.17.1. The dielectric withstanding voltage test shall be in accordance with 3.6.8 nd made at the same points as used for insulation resistance measurements. onremovable contact connectors shall have their rear portion enclosed for this test see 3.6.25)

4 6 27 Contact resistance The contact resistance shall be measured in accordance ith the contact resistance test of MIL-C-39029 (see 3.6 26).

4.6.28 Cover chain tensile strength. When applicable, the protective cover shall e securely held and a tensile static load of 25 pounds shall be applied to the end f the chain for 5 minutes in each direction as specified (see 3.6.27).

(a) With the axis of the chain at right angles to the axis of the holding fastener.
(b) With the axis of the chain in the same axis as that of the fastener.

4.6.29 Lanyard retention. The connector assembly shall be rigidly mounted as in ervice. The direction of force application shall be at an angle 15° from the onnector longitudinal axis. The force shall be applied at maximum rate of 50 pounds er second (see 3 6 28).

4.6.30 Lanyard release force (with the connector assembly mounted as in service). anyard release shall be accomplished by applying the force to the connector anyard at a rate of 10 to 20 pounds per second. Direction of pull shall be at any ngle within 15° of the connector longitudinal axis (see 3.6.29).

4 6 31 Coupling ring retention. Unmated connector plugs shall be suitably mounted o allow application of the force specified in table XII in a manner that would tend to eparate the coupling ring from the plug shell. The force shall be applied at a rate f approximately 100 pounds per minute and maintained for 10 seconds (see 3.6.30).

4 6 32 Low temperature, high humidity Wired mated connectors shall be placed in a est chamber and RH increased to 95 percent. The temperature shall be dropped to 55° ±2°C and held at that temperature for 3 hours While at low temperature at the end f the 3 hours, the connectors shall be tested to 4 6.30. After connector has eturned to 25° ±2°C and dried, it shall be examined for mechanical defects see 3.6.31).

4.6 33 Systems compatibility. Sample connectors and accessories representative of he products for which qualification is desired shall be inspected, tested, and valuated by the Government qualifying activity to determine the product's compatibility ith the system for which it is intended. Upon successful completion of the com- atibility evaluation, the qualifying activity shall issue a certification of compliance o the manufacturer (see 3 6 31).

4 6.33 1 Stores release The connector shall be correctly installed in the speci- ied aircraft and shall accept and release the applicable store with no evidence of aulty performance or malfunction (see 3 6 31 1).

5 PREPARATION FOR DELIVERY

5 1 Preparation for delivery shall be in accordance with MIL-C-55330.

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6. NOTES

6.1 Intended use.

(a) Individual connectors are intended for use where the operating temperature range of -55°C to 150°C is experienced Short time operation at higher temperatures is also possible (see figures 1 and 2).
(b) Adapter assemblies consist of a connector adapter with connectors attached to both ends. They are intended to be used in the temperature range from -55°C to 105°C. Short-time operation at higher temperatures is also possible (see figures 1 and 2).
(c) All connectors are environment resisting connectors with provisions to insure electrical continuity from plug to receptacle shell.
(d) Crimp contact connectors should have contacts installed in all positions when the connector is wired. Sealing plugs should be installed in the grommet holes when no wire is attached to the contact in grommet sealed connectors.
(e) The potting form should remain with the connector potting.
(f) Counterpart solder and crimp contact connectors are intended to be intermateable, but performance is then reduced to that of the solder contact connector.
(g) If air leakage requirements are critical, class P receptacle should be used and the connector should be potted.
(h) For finished wire diameters less than specified in table I, shrinkfit sleeving should be used over the wire.
(i) Where two or more wires are used in a solder cup or wire barrel moisture sealing is not obtainable without special provisions, the wire should be potted if sealing is required.

6.2 Ordering data.

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the complete part number and type (see 3.1 and 1.2).
(c) Levels of preservation-packaging, packing, and applicable marking (see 5.1).

6.2.1 Accessory hardware. Accessory hardware, such as covers, dummy stowage receptacles, or mounting hardware especially designed for these connectors, is defined in the applicable specification sheet. Flange gaskets should not be supplied with the receptacles unless they are specified (see 3 1).

6.3 Definitions. See MIL-STD-1353 and the following.

6.3.1 Gland clamp. A gland clamp, also called a cable sealing clamp, is a device intended to seal around a single-jacketed cable and is composed of a gland nut and sealing member.

6.3.2 Insulation spacer. An insulating spacer is a device intended to separate individual wires attached to the contacts of the connector.

6.3.3 Lanyard A loop of wire rope or other flexible material whereby a force may be applied to a plug connector to disengage it from the receptacle in a "break-away action"

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the application Qualified Products List (QPL) whether or not such products have actually been so listed by that date. The attention of suppliers is called to the requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6 4.1 Application for qualification The activity responsible for the QPL for MIL-C-81582 is the Naval Air Systems Command. The Naval Weapons Support Center has been designated by the Naval Air Systems Command as agent for establishment of the QPL Requests for information pertaining to and applications for qualification should be addressed to·

Commanding Officer
Naval Weapons Support Center
Crane, Indiana 47522

Attention Code 3074

6 5 Maximum recommended working voltage.

TABLE XX. Maximum recommended ac (rms) working voltage.

Condition	Service rating I	Service rating II
Sea level	600	1000
70,000 feet	300	450

6 6 Connector application information. The information shown on figure 8 is for guidance in design and application. The flashover voltages shown do not include corona or any safety factor.

6.7 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table XXI Table XXI shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible one with the other. Compatible couples between groups have been specified in table XXI based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table XXI shows, in addition to EMF against a calomel electrode, a derived "anodic index" with group 1 (gold, etc) as 0 and group 18 (magnesium, etc) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt

6 7.1 Groups Table XXI sets up 18 primary groups It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when couples with any member within the group, for example, such dissimilar metals as platinum and gold Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together

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TABLE XXI Compatible couples (see 6.7). 1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated, gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated, low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated, high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable, plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate, galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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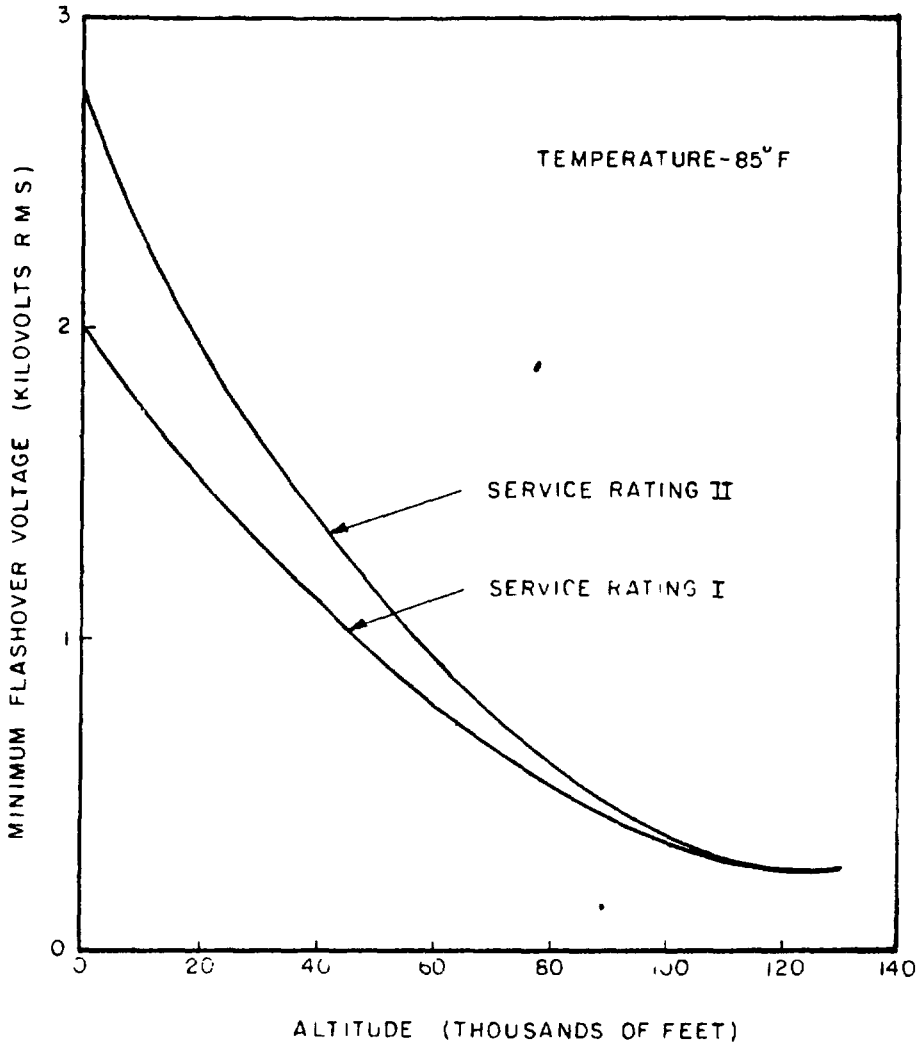


FIGURE 8. Typical flashover voltage values vs altitude on unmated miniature connectors

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6.7.2 Compatibility graphs. Permissible couple series are shown in table XXI by the graphs at the right. Members of groups connected by lines will form permissible couples. A "O" indicates the most cathodic member of each series, a "o" an anodic member, and the arrow indicates the anodic direction.

6.7.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table XXI. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part of the cathodic member of the couple, whenever practicable.

6.7.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table XXI, they are to be plated with those metals which will reduce the potential difference to that allowed by table XXI.

6.8 Information. Specification sheets should contain at least the following information unless specified herein

(a) Shell material
(b) Physical dimensions
(c) Weights
(d) Details for solder contacts and contacts for class H and thru-bulkhead connectors
(e) Connector identification and marking
(f) Connector polarization
(g) Applicable contact insertion and removal tools
(h) Temperature and service rating
(i) Special performance requirements

6.9 Patent notice. The Government has a royalty-free license under US Patent Application Serial No. 753,690 entitled "Releasable Electrical Connector" filed August 19, 1968 for the benefit of manufacturers of the items disclosed and described herein, either for the Government or for use in equipment delivered to the Government

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - EL
Navy - AS
Air Force - 80

Review activities:
Army - MU
Navy - EC
Air Force - 11, 15, 80
DSA - ES

User activity
Navy - SH

Preparing activity
Navy - AS

Agent
DSA - FS

(Project 5935-1953)

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